FCFDA: Fine–Coarse–Fine Progressive Graph Framework With Distribution Alignment for Hyperspectral Image Change Detection

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***Abstract*— Graph convolutional networks (GCNs) have attracted significant attention in hyperspectral image (HSI) change detection (CD) due to their capability to perform shape-adaptive convolutions and capture complex patterns within HSIs. Existing GCN-based methods typically preprocess bitemporal HSIs into graphs using a specific superpixel segmentation. However, this preprocessing step limits the modeling of spatial topologies to a fixed scale. Besides, these methods do not consider distribution shifts between bitemporal HSIs. To overcome these limitations, this article proposes a fine–coarse–fine progressive graph framework with distribution alignment (FCFDA) to learn progressive features across multilevel graphs for HSI-CD. Specifically, for each bitemporal HSI, we generate multiple hierarchical segmentations ranging from fine to coarse by gradually merging neighboring superpixels and subsequently transforming these segmentations into multilevel graphs. Second, instead of simply concatenating features from different hierarchies, FCFDA integrates them progressively from fine to coarse and then back to fine, generating subtle features tailored to the pixel-wise CD task. Finally, an effective distribution alignment (DA) method is designed to align the feature space of the bitemporal HSIs, thus mitigating the adverse effects of distribution shifts. Experiments conducted on real HSI-CD datasets demonstrate the effectiveness and superiority of the FCFDA.**

Received 29 July 2024; revised 10 October 2024; accepted 7 November 2024. Date of publication 14 November 2024; date of current version 27 November 2024. This work was supported in part by Zhejiang Provincial Natural Science Foundation of China under Grant LQ23F020012; in part by the National Natural Science Foundation of China under Grant 42171351, Grant 42122009, Grant 42471417, and Grant 62201272; in part by Zhejiang Province "Pioneering Soldier" and "Leading Goose" Research and Development Project under Grant 2023C01027; in part by the Science and Technology Development Fund, Macau, SAR, under Grant 0079/2023/ITP2; in part by the Wuyi University Hong Kong-Macau joint Research and Development Fund under Grant 2022WGALH16; and in part by China Scholarship Council. *(Corresponding authors: Shirui Pan; Jiangtao Peng.)*

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Digital Object Identifier 10.1109/TGRS.2024.3498034

***Index Terms*— Distribution alignment (DA), graph convolutional network (GCN), hyperspectral image (HSI) change detection (CD).**

I. Introduction

With advancements in spectral imaging technology, hyperspectral images (HSIs) now comprise an increasing number of spectral bands and capture finer spatial texture details in Earth observation [1], [2], [3], [4]. In the field of hyperspectral remote sensing, HSI change detection (CD) is a prevalent processing technique [5]. Its primary goal is to detect changed pixels by leveraging differences in spectral and spatial information across HSIs [6]. HSI-CD enables precise identification and comprehensive analysis of land cover, making it a competitive technology with various applications in environmental monitoring, urban planning, and agricultural assessment [7], [8], [9], [10].

Generally, HSI-CD methods can be classified into two categories: traditional methods and deep learning-based methods. Traditional methods mainly include algebra-based, transformation-based, and classification-based approaches [11]. Popular examples of these methods include change vector analysis (CVA) and multivariate change detection (MAD) [12], [13]. However, traditional methods require a heavy and challenging feature engineering stage. In deep learning-based HSI-CD methods, convolutional neural networks (CNNs) [14] have been widely applied due to their advantages of fewer parameters, shift invariance, and so on. For example, Wang et al. [15] designed an end-to-end 2-D CNN framework to explore abundant information on HSIs. Zhao et al. [16] proposed a simplified 3-D convolutional autoencoder to reduce spectral redundancy and improve the efficiency of HSI-CD. Moustafa et al. [17] introduced a recurrent residual UNet-based model to enhance CD performance. In [18], a global multihead interactive self-attention network was designed to explore the relevance and variation between different objects in bitemporal HSIs. Ou et al. [19] introduced slow feature analysis to select slow and fast spectral bands, which are fed into a CNN to extract more discriminative features for separating changed and unchanged pixels. Ding et al. [20] presented multiple spatial–spectral features aggregated neural networks, which leveraged multiscale and multilevel features from bitemporal HSIs to enhance temporal change information. Wu et al. [21] proposed combining HSI

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CD with band reweighting to enhance feature discrimination and improve CD accuracy. However, CNN-based methods still suffer from several limitations. First, these methods operate solely on data in Euclidean space, ignoring the non-Euclidean relationships and complex dependencies between hyperspectral pixels. Second, due to the complex and irregular shapes of objects in real-world HSIs, conventional CNNs with fixed convolutional kernels may struggle to effectively capture these irregularities and model long-range dependencies essential for accurate analysis.

In recent years, several studies have introduced graph neural networks (GNNs), such as graph convolutional networks (GCNs) [22] and graph attention networks [23], into HSI-CD, leveraging their powerful ability to model long-range spatial relationships in HSIs [24]. Existing GNN-based methods often utilize superpixel segmentation as preprocessing to reduce computational cost [25]. For example, Qu et al. [26] proposed a dual branch difference amplification GCN (D^2AGCN), which highlighted the differences between features of bitemporal HSIs. Wang et al. [27] presented a novel dual-branch framework based on temporal–spatial joint graph attention with complement strategy (CSDBF) to effectively exploit interaction information between bitemporal HSIs. Lin and Lin [28] introduced convex deep small-data learning theory into a GNN-based framework to ensure CD performance under a limited labeling rate. Zhang et al. [29] designed a multiorder GCN (MGCN) to leverage multiorder differential information between graph nodes. In [30], a sharpening-aware graph network was proposed to model dissimilarities between changed pixels and their neighbors and utilized Laplacian sharpening-based graph convolution to enhance these changes. Jian et al. [31] encoded data uncertainty and proposed an uncertainty-aware loss to promote reliable features playing a dominant role during feature learning. Ding et al. [32] proposed a graph representation learning-guided diffusion model to iteratively extract discriminative spectral–spatial features for high-precision CD. A graph transformer has also been introduced to model the long-term contextual relationships. For instance, Dong et al. [33] proposed a dual-branch local information-enhanced graph transformer (D-LIEG) to achieve efficient CD in complex scenarios. In [34], a graph transformer-guided multitemporal information united framework for HSI-CD (GTransCD) is proposed to capture global change information and integrate bitemporal information. In summary, the above methods achieve good CD performance by utilizing GNNs to model complex relationships within HSIs.

However, several challenges remain for further improvement. First, existing GNN-based methods operate only on single-scale graphs defined by presegmentation, limiting the effective utilization of spatial information. Details of the HSI information may be smoothed out in coarse-grained superpixels, while global information may not be effectively captured in fine-grained superpixels. As a result, CD performance exhibits significant limitations, such as sensitivity to superpixel segmentation results and the presence of false boundaries. Second, HSI-CD involves HSIs taken at different time points, where factors such as imaging time, climatic conditions, and so on can lead to distribution shifts between HSIs. Aligning bitemporal HSIs can minimize distribution shifts, enabling more accurate detection of real changes and enhancing the effectiveness and robustness of CD. However, this has not been well explored in existing methods.

To address the aforementioned challenges, this article proposes a fine–coarse–fine progressive graph framework with distribution alignment (FCFDA) for HSI-CD. First, hierarchical graph structures are constructed to model the subtle hierarchical relationships among scattered and irregular land covers. Specifically, we begin by dividing the HSIs into over-segmented superpixels and then generate a coarser superpixel segmentation with fewer superpixels by merging spatially adjacent superpixels. By iteratively repeating the superpixel merging process, we can progressively merge the initial over-segmented segmentation into a series of hierarchical superpixel segmentations, ranging from fine to coarse. Subsequently, each segmentation is transformed into a graph by creating edges between adjacent superpixels, with each superpixel represented as a node. Leveraging the merging relationships between hierarchical superpixels, we can seamlessly transfer features between any two neighboring hierarchical graphs. This helps to integrate different hierarchical graphs into a unified framework and enables them to work collaboratively.

Second, we extract subtle features through a fine-to-coarse (FTC) stage followed by a coarse-to-fine (CTF) stage. During the FTC stage, at each hierarchical level, graph representations are extracted by performing graph convolution and then converted to coarser-level features, which serve as the input for the next fine-to-coarse hierarchy. This enables the extraction of a series of hierarchical feature representations, spanning from fine to coarse. During the CTF stage, the hierarchical feature representations obtained from the FTC stage are progressively fused in a CTF manner. At each hierarchical level of the CTF stage, feature representations are extracted by performing graph convolution and then converted to finer-level features, which are then input into the next CTF hierarchy. Moreover, skip connections from the same hierarchy of the FTC stage are utilized to ensure that information is not lost due to scale changes.

Finally, to alleviate the distribution shifts between bitemporal HSIs and thus improve HSI-CD performance, we propose aligning the distributions of the bitemporal HSIs using a contrastive learning-based distribution alignment (DA) loss. Besides, an alternating optimization algorithm is proposed to optimize the parameters of the fine–coarse–fine progressive graph framework (FCF) and DA alternately and collaboratively.

The primary contributions are outlined as follows.

1) A novel FCFDA is proposed for highly precise HSI-CD by fully mining the spatial information of bitemporal HSIs across different hierarchies and effectively mitigating the challenges and adverse effects caused by distribution shifts. The FCFDA demonstrates superior performance on real HSI-CD datasets.

2) An FCF is built to model the spatial topologies of bitemporal HSIs in a fine–coarse–fine progressive hierarchical manner. Benefiting from the fine–coarse–fine strategy, subtle

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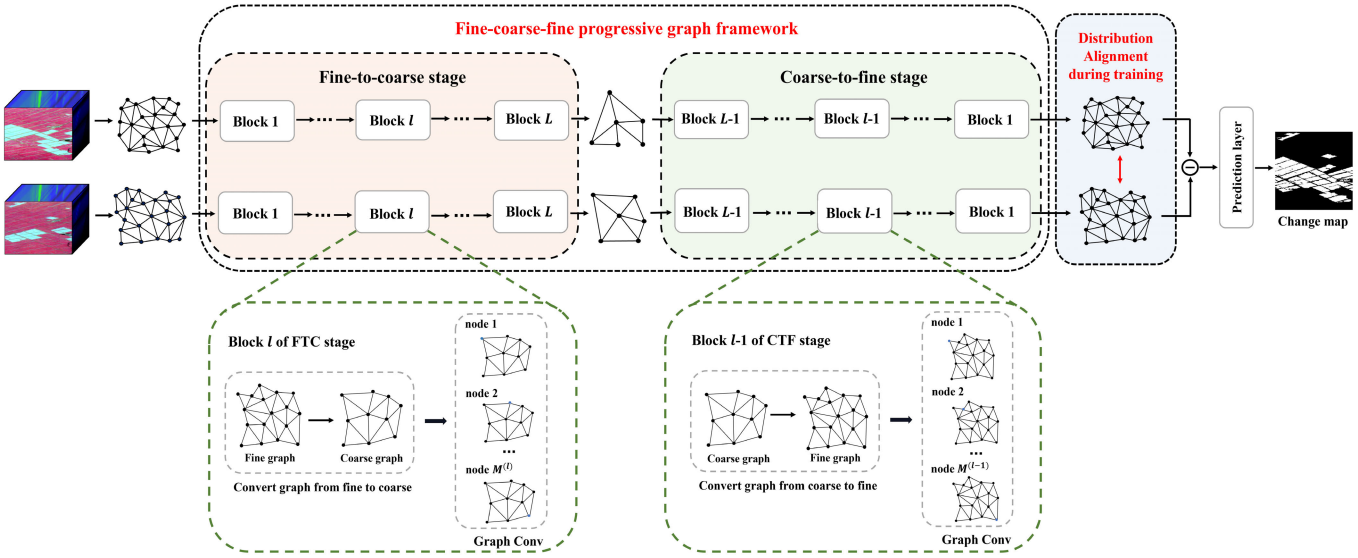


Fig. 1. Flowchart of the proposed FCFDA.

discriminative features can be learned, facilitating the differentiation between changed and unchanged pixels.

3) A contrastive learning-based DA loss and an effective alternating optimization algorithm are designed to align the distributions between bitemporal HSIs. This DA method significantly reduces false detections due to unrealistic changes and improves CD accuracy.

The remainder of this article is structured as follows. Section II presents the details of the proposed FCFDA. In Section III, a thorough and extensive performance evaluation is conducted on real HSI-CD datasets compared to various state-of-the-art methods. Finally, Section IV concludes this article.

II. Proposed Method

As shown in the flowchart of FCFDA in Fig. 1, the FCFDA first converts the bitemporal HSIs into two graphs, respectively. Then, the FCF is utilized to extract discriminative subtle features from fine to coarse and then back to fine, tailored for the pixel-wise HSI-CD task. Next, during training, DA is conducted to align the feature space of the bitemporal HSIs to mitigate the adverse effects of distribution shifts. Finally, the extracted features are fed into the prediction layer to generate CD results.

The proposed FCFDA is detailed by presenting the FCF (Sections II-B and II-C), describing the DA (Section II-D), elaborating the FCFDA-based CD (Section II-E), and introducing the alternating optimization algorithm (Section II-F).

A. Notations

The notations used in this article are as follows. The bitemporal HSIs are denoted by $\boldsymbol{Z}_{\mathcal{T}_1} \in \mathbb{R}^{H\times W\times B}$ and $\boldsymbol{Z}_{\mathcal{T}_2} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B are the height, width, and number of spectral bands, respectively. The set of labels is denoted by $\mathcal{C} = \{0, 1\}$, where 1 indicates a changed pixel and 0 indicates an unchanged pixel. The labeled sample set, consisting of N labeled pixels, is denoted by $\mathcal{D}_L = \{(\boldsymbol{z}_{\mathcal{T}_1,l_i}, \boldsymbol{z}_{\mathcal{T}_2,l_i}, y_i)\}_{i=1}^{N}$, where $\boldsymbol{z}_{\mathcal{T}_1,l_i} \in \mathbb{R}^B$ and $\boldsymbol{z}_{\mathcal{T}_2,l_i} \in \mathbb{R}^B$ are the spectral features of the ith labeled pixel in $\boldsymbol{Z}_{\mathcal{T}_1}$ and $\boldsymbol{Z}_{\mathcal{T}_2}$, respectively, and $y_i \in \mathcal{C}$ is its label.

B. Hierarchical Graph Construction

To fully exploit the spatial topologies of bitemporal HSIs, we convert them into a series of hierarchical graphs. Specifically, we initially segment each of the bitemporal HSIs into numerous superpixels and construct an initial graph. We then progressively merge spatially adjacent superpixels to generate graphs with progressive hierarchies, as illustrated in Fig. 2. The same hierarchical graph construction method is used for both bitemporal HSIs. Therefore, we will detail this process using one of the bitemporal HSIs as an example.

1) Construct Initial Graph: Given one of the bitemporal HSIs $\boldsymbol{Z}$ ($\boldsymbol{Z}_{\mathcal{T}_1}$ or $\boldsymbol{Z}_{\mathcal{T}_2}$), we consider it as a four-connected graph and use the superpixel hierarchy (SH) algorithm [35] to divide it into $M^{(1)}$ superpixels. Each superpixel is represented by its center feature, calculated by averaging the spectral signatures of the pixels within that superpixel, resulting in a feature matrix $\hat{\boldsymbol{X}} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{M^{(1)}}]^{\mathrm{T}} \in \mathbb{R}^{M^{(1)}\times B}$. Before constructing the initial graph, we transform $\hat{\boldsymbol{X}}$ into $\boldsymbol{X} = \hat{\boldsymbol{X}}\boldsymbol{W}_{\mathrm{pre}} \in \mathbb{R}^{M^{(1)}\times D}$ using a trainable weight $\boldsymbol{W}_{\mathrm{pre}} \in \mathbb{R}^{B\times D}$, where D denotes the dimension of the feature transformation.

Then, these superpixels are regarded as nodes to construct the initial graph $\mathcal{G}^{(1)} = (\mathcal{V}^{(1)}, \mathcal{E}^{(1)})$. $\mathcal{V}^{(1)}$ is represented by the feature matrix $\boldsymbol{X} \in \mathbb{R}^{M^{(1)}\times D}$, where the ith row of $\boldsymbol{X}$ denotes the feature of node $v_i^{(1)}$. $\mathcal{E}^{(1)}$ is represented by an adjacency matrix $\boldsymbol{A}^{(1)} \in \mathbb{R}^{M^{(1)}\times M^{(1)}}$, where $\boldsymbol{A}_{ij}^{(1)}$ indicates the presence of an edge between nodes v_i and v_j, which is determined by

$$\boldsymbol{A}_{ij}^{(1)} = \begin{cases} 1, & \text{if } v_i^{(1)} \text{ and } v_j^{(1)} \text{ are spatially adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $v_i^{(1)}$ and $v_j^{(1)}$ are considered spatially adjacent if they share a common boundary.

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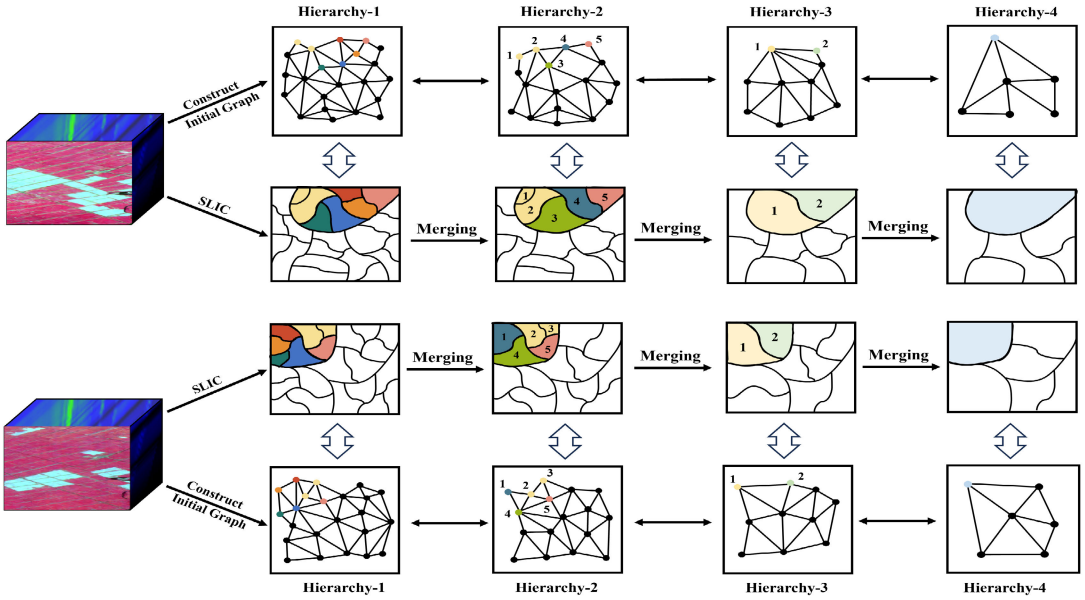


Fig. 2. Process of constructing hierarchical graphs.

2) Generate Hierarchical Graphs: Starting from the initial graph $\mathcal{G}^{(1)} = (\mathcal{V}^{(1)}, \mathcal{E}^{(1)})$, we utilize the SH algorithm to merge adjacent superpixels and obtain a superpixel segmentation $\mathcal{S}^{(2)} = \{\mathcal{C}_i^{(2)}\}_{i=1}^{M^{(2)}}$ with $M^{(2)}$ superpixels, where $M^{(2)} < M^{(1)}$. Based on $\mathcal{S}^{(2)}$, we generate the graph $\mathcal{G}^{(2)} = (\mathcal{V}^{(2)}, \mathcal{E}^{(2)})$ using the method described in Section II-B1.

Similarly, starting from the graph $\mathcal{G}^{(2)} = (\mathcal{V}^{(2)}, \mathcal{E}^{(2)})$, we continue to merge adjacent superpixel nodes to obtain a new superpixel segmentation $\mathcal{S}^{(3)} = \{\mathcal{C}_i^{(3)}\}_{i=1}^{M^{(3)}}$ with $M^{(3)}$ superpixels, where $M^{(3)} < M^{(2)}$. Based on $\mathcal{S}^{(3)}$, we generate the graph $\mathcal{G}^{(3)} = (\mathcal{V}^{(3)}, \mathcal{E}^{(3)})$.

By repeating the above process, we obtain a series of progressive superpixel segmentations and their corresponding graphs, denoted by $\{\mathcal{S}^{(l)}\}_{l=1}^{L}$ and $\{\mathcal{G}^{(l)}\}_{l=1}^{L}$, where L is a hyperparameter, $\mathcal{S}^{(l)} = \{\mathcal{C}_i^{(l)}\}_{i=1}^{M^{(l)}}$ represents the lth segmentation, consisting of $M^{(l)}$ superpixels, and $\mathcal{G}^{(l)} = (\mathcal{V}^{(l)}, \mathcal{E}^{(l)})$ represents the graph constructed from $\mathcal{S}^{(l)}$. Here, $M^{(1)} > M^{(2)} > \cdots > M^{(L)}$.

3) Transformation Between Hierarchical Graphs: According to the progressive merging relations between hierarchical superpixels, where any coarse-hierarchy superpixel segmentation is obtained by merging several smaller superpixels, we can easily perform feature transformations between different hierarchical graphs.

Let $\boldsymbol{Q}^{(l)} \in \mathbb{R}^{M^{(l-1)} \times M^{(l)}} (2 \le l \le L)$ denote the association matrix between $\mathcal{S}^{(l-1)}$ and $\mathcal{S}^{(l)}$, which is given as

$$\boldsymbol{Q}_{i,j}^{(l)} = \begin{cases} 1, & \text{if } \mathcal{C}_i^{(l-1)} \in \mathcal{C}_j^{(l)} \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Then, the features $\boldsymbol{H}_{\mathrm{F}}^{(l-1)}$ at a fine hierarchy of $\mathcal{G}^{(l-1)}$ can be transformed into the features $\boldsymbol{H}_{\mathrm{C}}^{(l)}$ at a coarser hierarchy of $\mathcal{G}^{(l)}$ by

$$\boldsymbol{H}_{\mathrm{C}}^{(l)} = \left(\hat{\boldsymbol{Q}}^{(l)}\right)^{\mathrm{T}} \boldsymbol{H}_{\mathrm{F}}^{(l-1)} \tag{3}$$

where $l = 2, 3, \ldots, L$, and $\hat{\boldsymbol{Q}}^{(l)}$ is the column-normalized version of $\hat{\boldsymbol{Q}}$, obtained by $\hat{\boldsymbol{Q}}_{i,j}^{(l)} = \boldsymbol{Q}_{i,j}^{(l)} / \sum_m \boldsymbol{Q}_{m,j}^{(l)}$. Specially, let $\boldsymbol{Q}^{(1)} \in \mathbb{R}^{HW \times M^{(1)}}$ denote the association matrix between pixels and $\mathcal{G}^{(1)}$.

In contrast, the features $\boldsymbol{H}_{\mathrm{C}}^{(l+1)}$ at a coarse hierarchy of $\mathcal{G}^{(l+1)}$ can be transformed into the features $\hat{\boldsymbol{H}}_{\mathrm{F}}^{(l)}$ at a finer hierarchy of $\mathcal{G}^{(l)}$ by

$$\hat{\boldsymbol{H}}_{\mathrm{F}}^{(l)} = \boldsymbol{Q}^{(l+1)} \boldsymbol{H}_{\mathrm{C}}^{(l+1)} \tag{4}$$

where $l = L-1, L-2, \ldots, 1$.

Utilizing the above transformations, feature conversion between different hierarchical levels of any two adjacent graphs can be flexibly achieved.

C. Fine–Coarse–Fine Progressive Graph Framework

To fully exploit the spatial topologies of HSIs on multilevel graphs, we propose an FCF. Specifically, the HSI is transferred to different hierarchical graphs to learn features at various levels.

1) Fine-to-Coarse Stage: Assuming that the FTC stage contains L blocks, then the feature extraction process is described as

$$\begin{cases} \bar{\boldsymbol{H}}^{(l)} = \mathrm{BN}\left(\left(\hat{\boldsymbol{Q}}^{(l)}\right)^{\mathrm{T}} \boldsymbol{H}^{(l-1)}\right) \\ \boldsymbol{H}^{(l)} = \varphi\left(\hat{\boldsymbol{A}}^{(l)} \bar{\boldsymbol{H}}^{(l)} \boldsymbol{W}_{\mathrm{FC}}^{(l)}\right) \end{cases} \tag{5}$$

where $l = 1, 2, \ldots, L$, $\boldsymbol{W}_{\mathrm{FC}}^{(l)} \in \mathbb{R}^{(D/2^{l-1}) \times (D/2^{l})}$ is the learnable matrix, $\mathrm{BN}(\cdot)$ denotes the batch normalization, and $\varphi(\cdot)$ is the leaky rectified linear unit (leaky ReLU). In particular, we set $\boldsymbol{H}^{(0)} = \boldsymbol{X}$. Besides, $\hat{\boldsymbol{A}}^{(l)}$ is the normalized version of the learned adjacency matrix of $\mathcal{G}^{(l)}$ and is formulated as

$$\hat{\boldsymbol{A}}^{(l)} = \left(\hat{\boldsymbol{D}}^{(l)}\right)^{-\frac{1}{2}} \left(\boldsymbol{A}_{\mathrm{ada}}^{(l)} + \boldsymbol{I}\right) \left(\hat{\boldsymbol{D}}^{(l)}\right)^{-\frac{1}{2}} \tag{6}$$

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where $\hat{\boldsymbol{D}}^{(l)}$ is the degree matrix of $\boldsymbol{A}_{\text{ada}}^{(l)}+\boldsymbol{I}$. Note that, instead of using the fixed graph $\boldsymbol{A}^{(l)}$, the adaptive graph $\boldsymbol{A}_{\text{ada}}^{(l)}$ is learned and used in each layer of the FCFDA, which is obtained by

$$\boldsymbol{A}_{\text{ada}}^{(l)}=\psi\left(\left(\bar{\boldsymbol{H}}^{(l)}\boldsymbol{W}_{\text{ada}}^{(l)}\right)\left(\bar{\boldsymbol{H}}^{(l)}\boldsymbol{W}_{\text{ada}}^{(l)}\right)^{\mathrm{T}}\right)\odot\boldsymbol{A}^{(l)} \tag{7}$$

where $\boldsymbol{W}_{\text{ada}}^{(l)}$ is a learnable matrix, $\odot$ represents the Hadamard product, and $\psi(\cdot)$ is the sigmoid function.

The FTC can learn different-level features from hierarchical graphs, which enables a more comprehensive exploitation of the spatial structures in HSIs.

2) CTF Stage: The extracted different-level features are progressively fused in a CTF manner during the CTF stage of FCFDA. Let $\boldsymbol{F}^{(l)}$ denote the extracted features of the lth block of the CTF stage and then we have

$$\begin{cases}\bar{\boldsymbol{F}}^{(l)}=\boldsymbol{Q}^{(l+1)}\boldsymbol{F}^{(l+1)}\\ \hat{\boldsymbol{F}}^{(l)}=\text{BN}\left(\text{concate}\left(\bar{\boldsymbol{F}}^{(l)},\boldsymbol{H}^{(l)}\right)\right)\\ \boldsymbol{F}^{(l)}=\varphi\left(\hat{\boldsymbol{A}}^{(l)}\hat{\boldsymbol{F}}^{(l)}\boldsymbol{W}_{\text{CF}}^{(l)}\right)\end{cases} \tag{8}$$

where $l=L-1,L-2,\ldots,1$, $\boldsymbol{W}_{\text{CF}}^{(l)}\in\mathbb{R}^{(D/2^{l+1}+2^l)\times(D/2^l)}$ is the learnable matrix, and concate$(\cdot)$ denotes concatenating features along the channel dimension. Particularly, $\boldsymbol{F}^{(L)}=\boldsymbol{H}^{(L)}$.

Finally, we transform $\boldsymbol{F}^{(1)}$ back into the HSI space and concatenate it with $\boldsymbol{X}$ to obtain the pixel-level output of the FCFDA using $\boldsymbol{F}=\text{concate}(\boldsymbol{Q}^{(1)}\boldsymbol{F}^{(1)},\boldsymbol{X})$.

During the CTF stage, different hierarchical features learned in the FTC stage are successively fused from coarse to fine, finally outputting the pixel-level feature map $\boldsymbol{F}$. It is important to note that, unlike the direct concatenation of different hierarchical features, the progressive fusion approach demonstrates a high degree of compatibility with the pixel-wise HSI-CD task.

For the sake of narrative convenience, the proposed FCFDA is briefly denoted as

$$\boldsymbol{F}\triangleq\text{FCFDA}(\boldsymbol{X},L) \tag{9}$$

where $\boldsymbol{X}$ is the input of the FCFDA. The procedures of the FCFDA are summarized in Algorithm 1.

D. Distribution Alignment

Ideally, unchanged objects should have similar spectral signatures across different temporal HSIs. However, due to varying imaging conditions, noticeable distribution shifts often occur, causing spectral differences in unchanged objects and potentially leading to false CDs.

To mitigate these effects, we propose aligning the feature space of bitemporal HSIs through a novel contrastive learning-based DA loss, thereby reducing the impact of such shifts. To this end, we consider the unchanged objects: if a pixel is unchanged, its features extracted by the FCFDA from the bitemporal HSIs should be as similar as possible.

Let $\mathcal{F}_u=\{(\boldsymbol{f}_{1,u_i},\boldsymbol{f}_{2,u_i})\}_{i=1}^{N_u}$ denote the set consisting of feature pairs of N_u unchanged pixels. For each unchanged pixel $\boldsymbol{f}_{1,u_i}$, we consider its dual feature $\boldsymbol{f}_{2,u_i}$ as the positive sample, while $\mathcal{N}(i)=\{\boldsymbol{f}_{2,u_j}|j\neq i\}$ are considered as the negative sample sets. Then, an unchanged DA loss is designed to make each unchanged pixel closer to its positive sample in the feature space while keeping it far away from its negative samples. The unchanged DA loss is formulated as follows:

$$\mathcal{L}_{\text{uDA}}=-\frac{1}{N_u}\sum_{i=1}^{N_u}\log\frac{\exp\left(\boldsymbol{f}_{1,u_i}^{\mathrm{T}}\boldsymbol{f}_{2,u_i}/\tau\right)}{\sum_{j\in\mathcal{N}(i)}\exp\left(\boldsymbol{f}_{1,u_i}^{\mathrm{T}}\boldsymbol{f}_{2,u_j}/\tau\right)} \tag{10}$$

where τ represents the temperature parameter. In the experiments, we set $\tau=0.5$.

On the other hand, let $\mathcal{F}_c=\{(\boldsymbol{f}_{1,c_i},\boldsymbol{f}_{2,c_i})\}_{i=1}^{N_c}$ denote the set consisting of feature pairs of N_c changed pixels. For each changed pixel $\boldsymbol{f}_{1,c_i}$, we consider its dual feature $\boldsymbol{f}_{2,c_i}$ as the negative sample. Then, we add a small disturbance to $\boldsymbol{f}_{1,c_i}$ to obtain the positive sample $\bar{\boldsymbol{f}}_{1,c_i}$. Similarly, a changed DA loss is introduced to enlarge the feature differences compared to negative samples, which is formulated as

$$\mathcal{L}_{\text{cDA}}=-\frac{1}{N_c}\sum_{i=1}^{N_c}\log\frac{\exp\left(\boldsymbol{f}_{1,c_i}^{\mathrm{T}}\bar{\boldsymbol{f}}_{1,c_i}/\tau\right)}{\exp\left(\boldsymbol{f}_{1,c_i}^{\mathrm{T}}\bar{\boldsymbol{f}}_{1,c_i}/\tau\right)+\exp\left(\boldsymbol{f}_{1,c_i}^{\mathrm{T}}\boldsymbol{f}_{2,c_i}/\tau\right)}. \tag{11}$$

Finally, the contrastive learning-based DA loss is given as follows:

$$\mathcal{L}_{\text{DA}}(\boldsymbol{W}_{\text{DA}})=\mathcal{L}_{\text{uDA}}+\mathcal{L}_{\text{cDA}} \tag{12}$$

where $\boldsymbol{W}_{\text{DA}}=\{\boldsymbol{W}_{\text{pre}},\{\boldsymbol{W}_{\text{FC}}^{(l)}\}_{l=1}^{L},\{\boldsymbol{W}_{\text{ada}}^{(l)}\}_{l=1}^{L},\{\boldsymbol{W}_{\text{CF}}^{(l)}\}_{l=1}^{L-1}\}$ represents the parameters of the FCF.

Algorithm 1 FCFDA

Input: Input $\boldsymbol{X}$, association matrixes $\{Q^{(l)}\}_{l=1}^{L}$ between hierarchical graphs.
Output: $\boldsymbol{F}$.

$\boldsymbol{H}^{(0)}=\boldsymbol{X}$;
for $l=1,2,\cdots,L$ **do**
  $\bar{\boldsymbol{H}}^{(l)}=\text{BN}((\hat{\boldsymbol{Q}}^{(l)})^{\mathrm{T}}\boldsymbol{H}^{(l-1)})$;
  $\boldsymbol{A}_{\text{ada}}^{(l)}=\psi((\bar{\boldsymbol{H}}^{(l)}\boldsymbol{W}_{\text{ada}}^{(l)})(\bar{\boldsymbol{H}}^{(l)}\boldsymbol{W}_{\text{ada}}^{(l)})^{\mathrm{T}})\odot\boldsymbol{A}^{(l)}$;
  $\hat{\boldsymbol{A}}^{(l)}=(\hat{\boldsymbol{D}}^{(l)})^{-\frac{1}{2}}(\boldsymbol{A}_{\text{ada}}^{(l)}+\boldsymbol{I})(\hat{\boldsymbol{D}}^{(l)})^{-\frac{1}{2}}$;
  $\boldsymbol{H}^{(l)}=\varphi(\hat{\boldsymbol{A}}^{(l)}\bar{\boldsymbol{H}}^{(l)}\boldsymbol{W}_{\text{FC}}^{(l)})$;
end
$\boldsymbol{F}^{(L)}=\boldsymbol{H}^{(L)}$;
for $l=L-1,L-2,\cdots,1$ **do**
  $\bar{\boldsymbol{F}}^{(l)}=\boldsymbol{Q}^{(l+1)}\boldsymbol{F}^{(l+1)}$;
  $\hat{\boldsymbol{F}}^{(l)}=\text{BN}(\text{concate}(\bar{\boldsymbol{F}}^{(l)},\boldsymbol{H}^{(l)}))$;
  $\boldsymbol{F}^{(l)}=\varphi(\hat{\boldsymbol{A}}^{(l)}\hat{\boldsymbol{F}}^{(l)}\boldsymbol{W}_{\text{CF}}^{(l)})$;
end
return $\boldsymbol{F}=concate(\boldsymbol{Q}^{(1)}\boldsymbol{F}^{(1)},\boldsymbol{X})$.

E. FCFDA-Based CD

First, given the bitemporal HSIs $\boldsymbol{Z}_{\mathcal{T}_1}$ and $\boldsymbol{Z}_{\mathcal{T}_2}$, we transform them into the graph space and obtain the initial superpixel features $\boldsymbol{X}_{\mathcal{T}_1}\in\mathbb{R}^{M_1^{(1)}\times D}$ and $\boldsymbol{X}_{\mathcal{T}_2}\in\mathbb{R}^{M_2^{(1)}\times D}$ using the method described in Section II-B1, where $M_1^{(1)}$ and $M_2^{(1)}$ represent the number of initial superpixels for the bitemporal HSIs.

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Second, we generate hierarchical graphs $\{\mathcal{G}_{\mathcal{T}_1}^{(l)}\}_{l=1}^{L}$ and $\{\mathcal{G}_{\mathcal{T}_2}^{(l)}\}_{l=1}^{L}$ for the bitemporal HSIs using the method described in Section II-B2. These hierarchical graphs can fully model the subtle spatial topologies of the bitemporal HSIs from fine to coarse.

Next, given the inputs $\boldsymbol{X}_{\mathcal{T}_1}$ and $\boldsymbol{X}_{\mathcal{T}_2}$, the FCFDA will output the extracted pixel-level feature representations of the bitemporal HSIs, which is formulated as

$$\begin{cases} \boldsymbol{F}_{\mathcal{T}_1} = \text{FCFDA}_{\mathcal{T}_1}\left(\boldsymbol{X}_{\mathcal{T}_1}, L\right) \\ \boldsymbol{F}_{\mathcal{T}_2} = \text{FCFDA}_{\mathcal{T}_2}\left(\boldsymbol{X}_{\mathcal{T}_2}, L\right). \end{cases} \tag{13}$$

The resulting $\boldsymbol{F}_{\mathcal{T}_1}$ and $\boldsymbol{F}_{\mathcal{T}_2}$ are then subtracted as

$$\boldsymbol{D}_{\text{sub}} = \text{abs}\left(\boldsymbol{F}_{\mathcal{T}_1} - \boldsymbol{F}_{\mathcal{T}_2}\right) \tag{14}$$

where $\text{abs}(\cdot)$ denotes taking the absolute values.

Finally, $\boldsymbol{D}_{\text{sub}}$ is fed into a prediction layer, which predicts the change possibility score as

$$\begin{cases} \bar{\boldsymbol{D}}_{\text{sub}} = \varphi(\text{BN}(\boldsymbol{D}_{\text{sub}}\boldsymbol{W}_{\text{e}})) \\ \hat{\boldsymbol{Y}} = \psi\left(\bar{\boldsymbol{D}}_{\text{sub}}\boldsymbol{W}_{\text{p}}\right) \end{cases} \tag{15}$$

where $\boldsymbol{W}_{\text{e}}$ and $\boldsymbol{W}_{\text{p}}$ are the trainable matrices in the prediction layer, and $\psi(\cdot)$ is the sigmoid function.

For each pixel, the FCFDA will generate a change possibility score and attempt to match it to the ground truth label, that is, 1 for changed and 0 for unchanged. Thus, the cross-entropy loss is utilized to optimize the parameters of FCFDA

$$\mathcal{L}_{\text{CE}}(\boldsymbol{W}) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(\hat{y}_i\right) + (1-y_i)\log\left(1-\hat{y}_i\right)\right] \tag{16}$$

where $\hat{y}_i$ and y_i are the predicted label and true label of the ith labeled pixel, respectively. $\boldsymbol{W} = \{\boldsymbol{W}_{\text{DA}}, \boldsymbol{W}_{\text{e}}, \boldsymbol{W}_{\text{p}}\}$ represents all the parameters of the FCFDA, which includes both the parameters of the FCF and the parameters of the prediction layer.

F. Alternating Optimization Algorithm

In this work, DA is integrated into the FCF to mitigate the adverse effects of distribution shifts. A straightforward approach would be to combine the DA loss and cross-entropy loss into a joint loss function. However, the objectives of DA and CD are not entirely the same. DA primarily aims to mitigate distribution shifts between bitemporal HSIs, while CD focuses on accurately capturing the changes. Consequently, combining these two objectives in a joint loss may lead to gradient conflicts or interference during the optimization. In addition, our experimental results showed that this joint loss had limited effectiveness in achieving proper DA (see Section III-F). Therefore, we propose an alternating optimization scheme to decouple these two objectives, optimizing DA and CD alternately. This strategy allows for more flexible handling of their distinct optimization requirements. Specifically, in each iteration, we first update all the parameters that contribute to the CD task and then only update the parameters related to DA. The detailed optimization process is described as follows.

Algorithm 2 Training Algorithm of the FCFDA

Input: The bitemporal HSIs $\boldsymbol{Z}_{\mathcal{T}_1}$ and $\boldsymbol{Z}_{\mathcal{T}_2}$, learning rate η and η_{DA}, the number of training iterations $T_{\max}$.
Output: The trained parameters $\boldsymbol{W}$.

1 Set $t = 1$, randomly initialize the trainable parameters;
2 Transform $\boldsymbol{Z}_{\mathcal{T}_1}$ and $\boldsymbol{Z}_{\mathcal{T}_2}$ into the graph space, and obtain the initial superpixel features $\boldsymbol{X}_{\mathcal{T}_1}$ and $\boldsymbol{X}_{\mathcal{T}_2}$ using the method in Section II-B1;
3 **repeat**
4 **// Change Detection.**
5 $\boldsymbol{F}_{\mathcal{T}_1} = \text{FCFDA}_{\mathcal{T}_1}(\boldsymbol{X}_{\mathcal{T}_1}, L)$;
6 $\boldsymbol{F}_{\mathcal{T}_2} = \text{FCFDA}_{\mathcal{T}_2}(\boldsymbol{X}_{\mathcal{T}_2}, L)$;
7 $\boldsymbol{D}_{\text{sub}} = \text{abs}(\boldsymbol{F}_{\mathcal{T}_1} - \boldsymbol{F}_{\mathcal{T}_2})$;
8 $\hat{\boldsymbol{Y}} = \psi(\varphi(\text{BN}(\boldsymbol{D}_{\text{sub}}\boldsymbol{W}_{\text{e}}))\boldsymbol{W}_{\text{p}})$;
9 Calculate the cross-entropy loss $\mathcal{L}_{\text{CE}}(\boldsymbol{W})$;
10 Update parameters: $\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta\frac{\partial \mathcal{L}_{\text{CE}}(\boldsymbol{W})}{\partial \boldsymbol{W}}$.
11 **// Distribution Alignment.**
12 **for** $i = 1$ to ν **do**
13 $\boldsymbol{F}_{\mathcal{T}_1} = \text{FCFDA}_{\mathcal{T}_1}(\boldsymbol{X}_{\mathcal{T}_1}, L)$;
14 $\boldsymbol{F}_{\mathcal{T}_2} = \text{FCFDA}_{\mathcal{T}_2}(\boldsymbol{X}_{\mathcal{T}_2}, L)$;
15 Calculate $\mathcal{L}_{\text{uDA}}$ and $\mathcal{L}_{\text{cDA}}$ by Eq.(10) and Eq.(11);
16 $\mathcal{L}_{\text{DA}}(\boldsymbol{W}_{\text{DA}}) = \mathcal{L}_{\text{uDA}} + \mathcal{L}_{\text{cDA}}$;
17 Update: $\boldsymbol{W}_{\text{DA}} \leftarrow \boldsymbol{W}_{\text{DA}} - \eta_{\text{DA}}\frac{\partial \mathcal{L}_{\text{DA}}(\boldsymbol{W}_{\text{DA}})}{\partial \boldsymbol{W}_{\text{DA}}}$.
18 **end**
19 $t \leftarrow t + 1$;
20 **until** $t > T_{\max}$;
21 **return** $\boldsymbol{W}$.

1) Update $\boldsymbol{W}$ Related to CD: When performing CD, we compute the cross-entropy loss and then update all the trainable parameters contained in the FCFDA, that is, $\boldsymbol{W} = \{\boldsymbol{W}_{\text{DA}}, \boldsymbol{W}_{\text{e}}, \boldsymbol{W}_{\text{p}}\}$, as follows:

$$\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta\frac{\partial \mathcal{L}_{\text{cel}}(\boldsymbol{W})}{\partial \boldsymbol{W}} \tag{17}$$

where η is the learning rate for CD.

2) Update $\boldsymbol{W}_{DA}$ Related to Distribution Alignment: For DA, ignoring the prediction process, we calculate the DA loss and only update the parameters related to DA, that is, $\boldsymbol{W}_{\text{DA}}$, as follows:

$$\boldsymbol{W}_{\text{DA}} \leftarrow \boldsymbol{W}_{\text{DA}} - \eta_{\text{DA}}\frac{\partial \mathcal{L}_{\text{DA}}(\boldsymbol{W}_{\text{DA}})}{\partial \boldsymbol{W}_{\text{DA}}} \tag{18}$$

where η_{DA} is the learning rate for the DA.

3) Training Algorithm: With the above updating rules, the alternating optimization algorithm for the FCFDA is shown in Algorithm 2. In line 1, all the trainable parameters of the FCFDA are randomly initialized. In line 2, the bitemporal HSIs are transformed into the graph space and then preprocessed to obtain the initial superpixel features. From lines 3 to 20, the CD parameters $\boldsymbol{W}$ and the DA parameters $\boldsymbol{W}_{\text{DA}}$ are updated alternatively and iteratively. Specifically, $\boldsymbol{W}$ is updated once in each iteration, while $\boldsymbol{W}_{\text{DA}}$ is updated ν times every iteration. Note that we set $\nu = 1$ in the experiments to ensure efficiency, while a larger ν would allow the DA to play a more significant role in feature learning.

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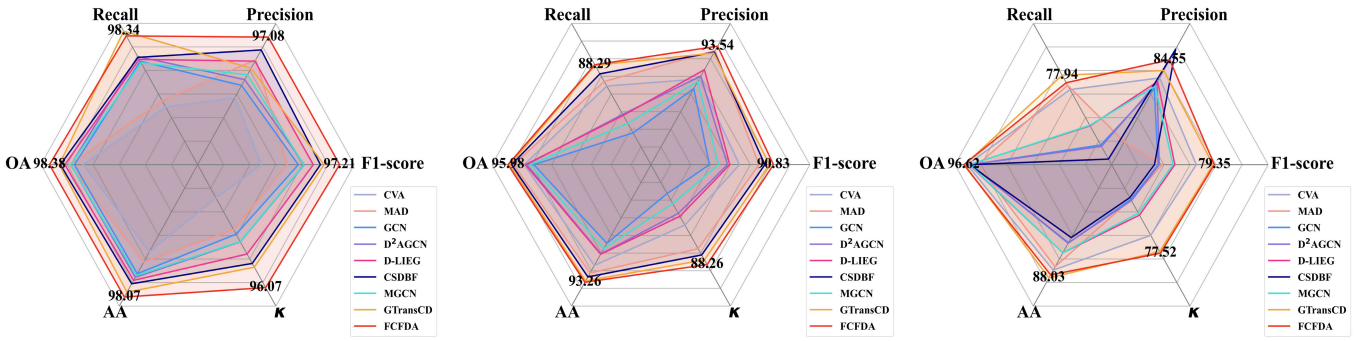


Fig. 3. Accuracy overview on real HSI-CD datasets. (a) Farmland. (b) USA. (c) River.

III. Experiment

A. Experimental Settings

1) Datasets: Experiments are evaluated on real HSI-CD datasets, including Farmland, River, and USA [36].

The Farmland dataset was collected over farmland near Yancheng City, Jiangsu province, China. Both bitemporal HSIs contain 450 × 140 pixels and 155 spectral bands, acquired on May 3, 2006, and April 23, 2007, respectively. The main change in this dataset is the type of farmland cover.

The River dataset was obtained in Jiangsu province, China. Both bitemporal HSIs contain 463 × 241 pixels and 198 spectral bands, acquired on May 3, 2013, and December 31, 2013, respectively. The main change in this dataset is the disappearance of substances in the river.

The USA dataset was obtained over an agricultural field in Hermiston City, USA. Both bitemporal HSIs contain 307 × 241 pixels and 154 spectral bands, acquired on May 1, 2004, and May 8, 2007, respectively. The main change in the USA dataset is the type of land cover.

2) Comparison Methods: To assess performance, the proposed FCFDA is compared against the CVA, the MAD, and also several state-of-the-art GNN-based methods, namely GCN, D^{2}AGCN (2022) [26], CSDBF (2022) [27], D-LIEG (2023) [33], MGCN (2024) [29], and GTransCD (2024) [34]. The GCN is implemented based on the framework of D^{2}AGCN, but contains only two graph convolutional layers without using other modules. In the experiments, these methods use their default hyperparameter configurations.

3) Parameter Setting: On all datasets, the parameter settings are consistent for our FCFDA. The dimension of the feature is set to $D = 128$. We consider $L = 4$ hierarchies in our fine–coarse–fine framework, with the numbers of superpixels in the four hierarchies set to [4096, 2048, 1024, 512]. The learning rate η is set to 0.005, and the learning rate η_{DA} is set to 0.1. The number of training iterations, $T_{\max}$, is set to 300. In the experiments, only 1% of labeled samples are randomly selected as the training set.

4) Evaluation Indicators: Overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), Precision, Recall, and F1-score are used as the main performance indices for quantitative evaluation [37]. For these performance indices, a larger value indicates superior CD performance. All indices are reported by averaging the results from ten random runs.

To visually depict the CD performance, change maps are illustrated. In these change maps, true-positive (TP) pixels, true-negative (TN) pixels, false-positive (FP) pixels, and false-negative (FN) pixels are shown in white, black, blue, and red, respectively. This visualization helps compare the advantages and disadvantages of different methods.

B. CD Performance

The overview of accuracy comparison is shown in Fig. 3. As can be seen, our FCFDA performs either on par with or better than the state-of-the-art methods in terms of all performance indexes on these real datasets. Detailed accuracy and visualization comparisons with these state-of-the-art methods will be expanded below.

1) Farmland Dataset: Table I reports the CD results of all compared methods on the Farmland dataset. The best CD accuracies are emphasized in boldface, while the second-place methods are highlighted with underlines. The experimental results indicate that CVA performs competitively on the Farmland dataset, particularly in terms of OA. The GCN exhibits relatively lower performance on this challenging dataset, as it is difficult to distinguish the changed pixels from the unchanged pixels without considering more information between the bitemporal HSIs. Compared to the basic GCN method, the D^{2}AGCN demonstrates better performance, indicating the effectiveness of enhancing the change information between bitemporal HSIs. By combining the advantages of the GNN and the CNN, the D-LIEG can model both local and global information well, resulting in significantly better OA, AA, and F1-score. The CSDBF achieves better OA and also demonstrates satisfactory accuracies in terms of other indicators. This indicates that interacting information between bitemporal HSIs improves the model's ability to detect changed and unchanged pixels. Benefitting from applying a transformer to the graph structure to capture local and global-range information, the GTransCD achieves suboptimal OA and significantly improves Recall. Compared to other GNN-based methods, the proposed FCFDA demonstrates superior performance in almost all evaluation indicators. For example, our FCFDA achieves the highest accuracy for the unchanged class and the second-highest accuracy for the changed class, demonstrating its advantage in detecting both changed and unchanged pixels. The best OA of 98.38%, nearly 1.87% higher than the second-place method, indicates that

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TABLE I
CD PERFORMANCE ON THE FARMLAND DATASET WITH 1% TRAINING SAMPLES

Method	Change(%)	Unchange(%)	OA(%)	AA(%)	$\kappa \times 100$	F1-score(%)	Precision(%)	Recall(%)
CVA	93.92±0.01	95.37±0.01	94.95±0.01	94.64±0.00	88.03±0.01	91.63±0.01	89.89±0.02	93.92±0.01
MAD	94.54±0.01	94.91±0.01	94.81±0.01	94.73±0.01	87.72±0.02	91.43±0.01	88.89±0.02	94.54±0.01
GCN	91.96±2.53	94.28±0.90	93.60±0.20	93.12±0.84	84.74±0.60	89.29±0.50	86.84±1.51	91.96±2.53
D^2AGCN	92.97±1.82	94.83±0.69	94.29±0.11	93.90±0.58	86.36±0.34	90.43±0.29	88.06±1.19	92.97±1.82
D-LIEG	92.36±1.14	96.71±0.29	95.44±0.14	94.53±0.43	88.95±0.40	92.16±0.31	91.98±0.57	92.36±1.14
CSDBF	92.79±3.12	97.72±0.46	96.29±0.70	95.26±1.40	90.94±1.82	93.53±1.36	94.36±0.95	92.79±3.12
MGCN	91.65±0.83	95.41±0.55	94.32±0.35	93.53±0.38	86.32±0.82	90.35±0.56	89.09±1.15	91.65±0.83
GTransCD	**98.34±0.62**	95.76±0.77	96.51±0.38	97.05±0.13	91.74±0.84	94.24±0.56	90.48±1.52	**98.34±0.62**
FCFDA	97.34±0.41	**98.80±0.15**	**98.38±0.17**	**98.07±0.22**	**96.07±0.40**	**97.21±0.29**	**97.08±0.36**	97.34±0.41

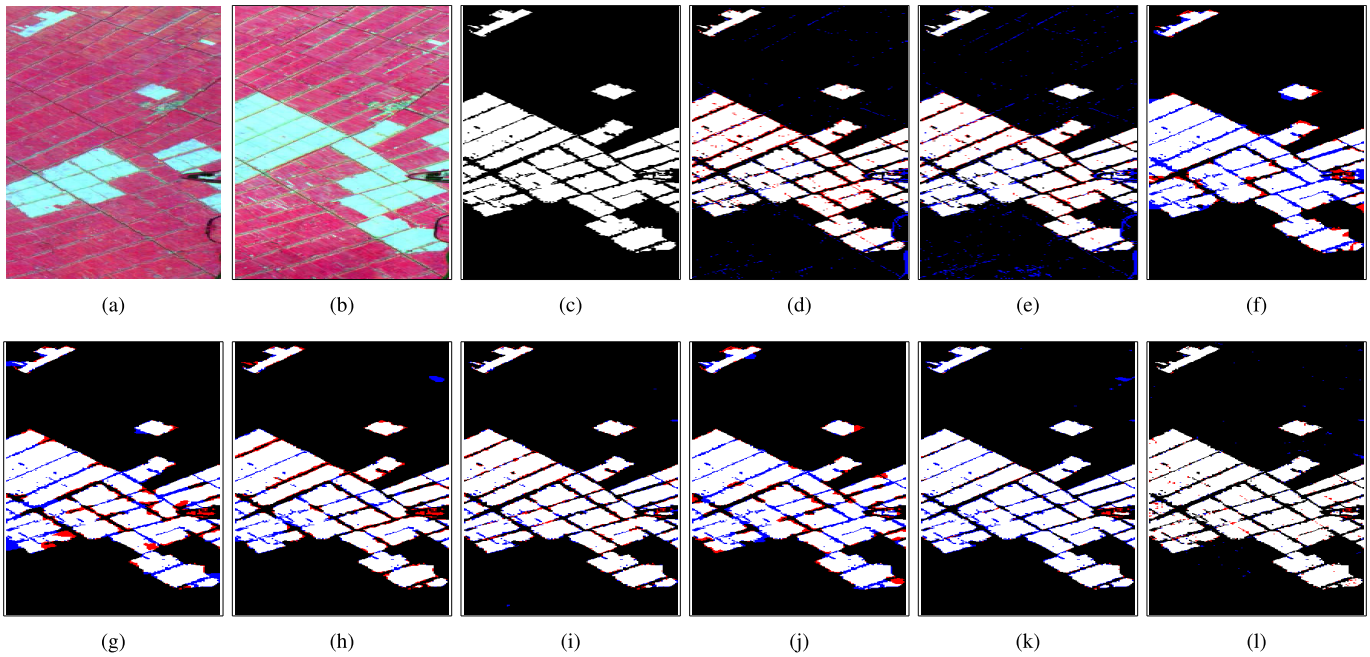


Fig. 4. CD maps of the compared methods on the Farmland dataset. White, black, blue, and red represent the TP pixels, TN pixels, FP pixels, and FN pixels, respectively. (a) False color image of Farmland dataset obtained in May 2006. (b) False color image of Farmland dataset obtained in April 2007. (c) Ground truth. (d) CVA. (e) MAD. (f) GCN. (g) D^2AGCN. (h) D-LIEG. (i) CSDBF. (j) MGCN. (k) GTransCD. (l) FCFDA.

the FCFDA correctly identifies nearly all samples. The best AA suggests that the FCFDA achieves good performance across different categories. The best κ and F1-score prove the overall effectiveness of the proposed FCFDA. These improvements can be attributed to the proposed FCF and the DA strategy. Besides, it is worth highlighting that the FCFDA has a significant advantage in Precision, being almost 2.72% higher than the second place. A possible reason is that, after aligning distribution shifts between bitemporal HSIs, the proposed FCFDA is capable of significantly reducing pseudo-changes. In general, our FCFDA outperforms these compared methods under the 1% training ratio condition, demonstrating its superiority.

Fig. 4 shows the visual CD maps on the Farmland dataset. GCN and the D^2AGCN exhibit a large number of misclassified pixels around the edges of the changed areas and small targets. After introducing the CNN to capture local spatial information, D-LIEG has fewer misclassified pixels than the GCN and the D^2AGCN, but still has many FP pixels around the edges. The CSDBF shows better performance in distinguishing changed and unchanged pixels but still produces many FP and FN pixels, as seen in the edge areas of the change map. The GTransCD shows better visual performance than the above methods but generates a lot of pseudo-changed pixels (blue ones). From the visual observations, the proposed FCFDA achieves the best visual performance, evidenced by fewer misclassified pixels and better details than the other compared methods.

2) USA Dataset: Table II displays the accuracies of all comparison methods on the USA dataset. It can be seen that CVA does not perform well, whereas MAD consistently outperforms CVA across all metrics on the USA dataset. The GCN, the D^2AGCN, the D-LIEG, and the MGCN perform poorly on this dataset, with none exceeding 92% in terms of OA. The CSDBF achieves improved CD accuracy, which can be attributed to its ability to model interaction

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TABLE II
CD PERFORMANCE ON THE USA DATASET WITH 1% TRAINING SAMPLES

Method	Change(%)	Unchange(%)	OA(%)	AA(%)	$\kappa \times 100$	F1-score(%)	Precision(%)	Recall(%)
CVA	82.25±0.01	94.72±0.01	91.91±0.01	88.49±0.01	77.07±0.02	82.26±0.02	84.27±0.03	82.25±0.01
MAD	83.55±0.03	97.76±0.01	94.55±0.01	90.65±0.01	83.76±0.02	87.19±0.01	92.29±0.01	83.55±0.03
GCN	68.99±2.55	95.45±0.51	89.48±0.36	82.22±1.09	68.14±1.41	74.71±1.26	81.56±1.30	68.99±2.55
D^2AGCN	74.24±2.72	96.21±0.19	91.25±0.55	85.22±1.31	73.75±1.92	79.26±1.61	85.06±0.56	74.24±2.72
D-LIEG	73.94±3.30	96.73±0.83	91.59±0.48	85.33±1.36	74.57±1.69	79.83±1.47	86.91±2.46	73.94±3.30
CSDBF	85.71±1.87	97.79±0.16	95.07±0.35	91.75±0.89	85.53±1.14	88.67±0.93	91.87±0.46	85.71±1.87
MGCN	71.68±4.57	95.82±1.17	90.38±0.36	83.75±1.74	70.97±1.64	77.00±1.58	83.53±3.35	71.68±4.57
GTransCD	88.14±1.81	97.63±0.65	95.49±0.22	92.89±0.63	86.92±0.58	89.81±0.45	91.60±1.93	88.14±1.81
FCFDA	**88.29±1.25**	**98.22±0.25**	**95.98±0.09**	**93.26±0.50**	**88.26±0.35**	**90.83±0.30**	**93.54±0.75**	**88.29±1.25**

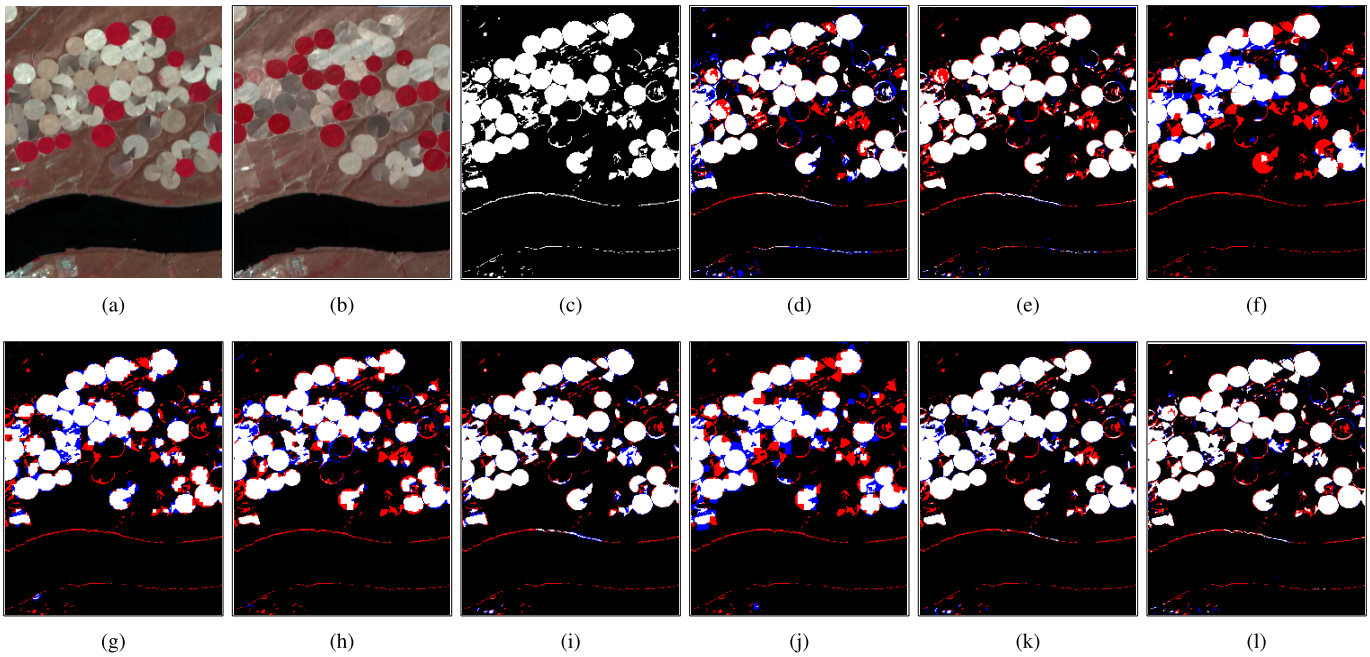


Fig. 5. CD maps of the compared methods on the USA dataset. White, black, blue, and red represent the TP pixels, TN pixels, FP pixels, and FN pixels, respectively. (a) False color image of USA dataset obtained in May 2004. (b) False color image of USA dataset obtained in May 2007. (c) Ground truth. (d) CVA. (e) MAD. (f) GCN. (g) D^2AGCN. (h) D-LIEG. (i) CSDBF. (j) MGCN. (k) GTransCD. (l) FCFDA.

information between bitemporal HSIs, enabling more effective discrimination between changed and unchanged pixels. The GTransCD greatly improves CD accuracies by effectively utilizing local and global-range spatial information among HSIs. The accuracy of FCFDA is also improved, as it can fully consider and adaptively fuse spatial information across different hierarchies, thereby extracting subtle discriminative features. Our FCFDA achieves the highest accuracy for both the changed and unchanged classes on the USA dataset. Most notably, the FCFDA shows significant advantages in the Precision index, achieving almost 1.67% improvement over the second-place CSDBF. This again demonstrates that DA helps filter out pseudo-changes and precisely identify changed regions of land covers. In summary, compared to all other methods, our FCFDA achieves superior performance across almost all indices.

Fig. 5 presents the visual CD maps of all comparison methods on the USA dataset. For the GCN, the D^2AGCN, the D-LIEG, and the MGCN, there are a large number of misclassified pixels (blue and red pixels) in their CD maps, which is consistent with the numerical results in Table II. The CSDBF shows improved visual performance but still produces many misclassified pixels around the edges between the changed and unchanged regions. Compared to these methods, the FCFDA and the GTransCD show significantly better visual performance. In addition, the FCFDA produces fewer FP pixels than the GTransCD, which can be attributed to the DA.

3) River Dataset: Table III presents the CD accuracies on the River dataset. Except for the GTransCD and the proposed FCFDA, the other methods perform similarly on this dataset and do not achieve satisfactory accuracies, especially in terms of κ, F1-score, and Recall indices. As shown in Table III, the proposed FCFDA and the GTransCD are superior to the other compared methods in κ and F1-score, proving the overall effectiveness of the GTransCD and our FCFDA. The proposed FCFDA also has clear advantages in Precision, achieving

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TABLE III
CD PERFORMANCE ON THE RIVER DATASET WITH 1% TRAINING SAMPLES

Method	Change(%)	Unchange(%)	OA(%)	AA(%)	$\kappa \times 100$	F1-score(%)	Precision(%)	Recall(%)
CVA	71.89±0.04	97.54±0.00	95.31±0.01	84.72±0.02	69.96±0.04	72.45±0.04	76.84±0.03	71.89±0.04
MAD	73.72±0.03	92.91±0.01	91.24±0.01	83.32±0.01	54.71±0.02	59.41±0.02	49.75±0.03	73.72±0.03
GCN	48.38±5.53	98.24±0.46	93.91±0.16	73.31±2.55	54.71±3.05	57.84±3.09	72.64±2.68	48.38±5.53
D^2AGCN	47.78±0.88	98.44±0.40	94.04±0.31	73.11±0.31	55.20±1.23	58.24±1.03	74.80±4.74	47.78±0.88
D-LIEG	56.52±6.15	98.19±0.53	94.57±0.17	77.35±2.82	61.38±3.00	64.24±3.05	75.29±3.94	56.52±6.15
CSDBF	61.13±12.66	98.64±0.52	95.38±0.64	79.88±6.08	66.67±7.83	69.07±7.64	81.62±3.14	61.13±12.66
MGCN	56.72±6.77	97.96±0.76	94.38±0.30	77.34±3.05	60.56±3.02	63.53±3.06	73.32±5.15	56.72±6.77
GTransCD	**77.94±3.71**	98.12±0.48	96.36±0.15	**88.03±1.62**	76.85±0.55	78.84±0.54	79.99±3.43	**77.94±3.71**
FCFDA	74.80±1.28	**98.69±0.21**	**96.62±0.08**	86.75±0.54	**77.52±0.15**	**79.35±0.10**	**84.55±1.85**	74.80±1.28

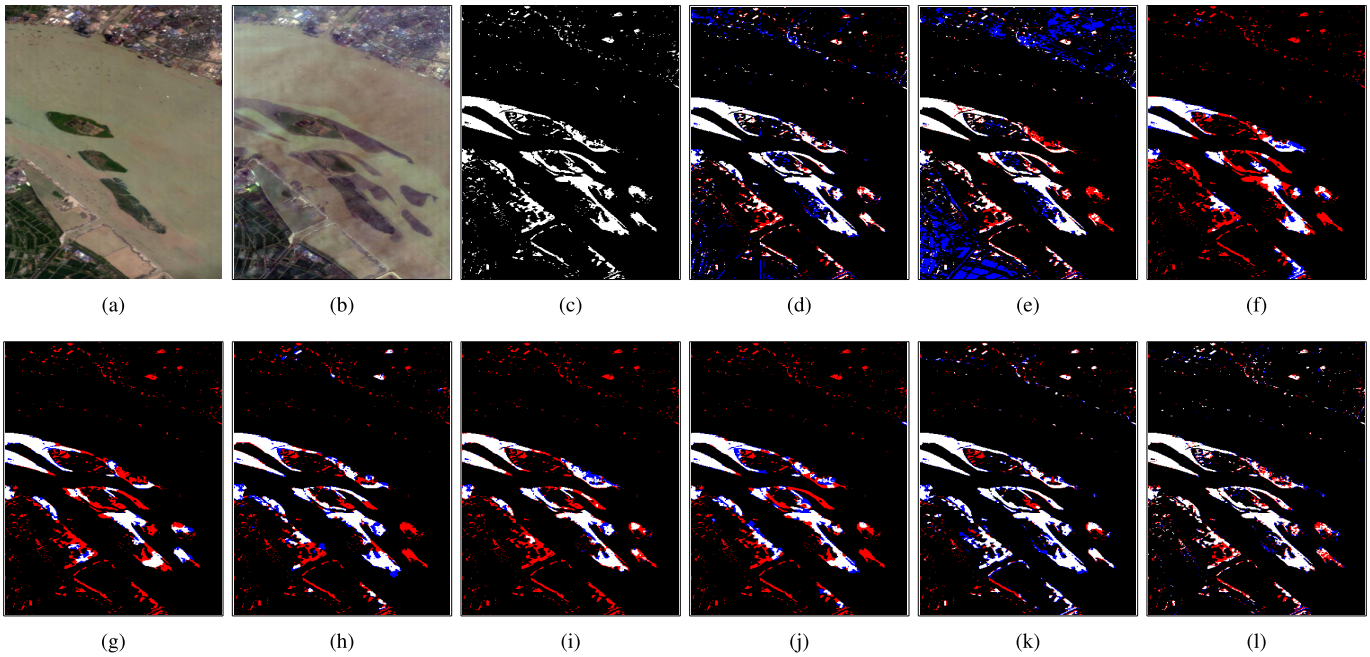


Fig. 6. CD maps of the compared methods on the River dataset. White, black, blue, and red represent the TP pixels, TN pixels, FP pixels, and FN pixels, respectively. (a) False color image of River dataset obtained in May 2013. (b) False color image of River dataset obtained in December 2013. (c) Ground truth. (d) CVA. (e) MAD. (f) GCN. (g) D^2AGCN. (h) D-LIEG. (i) CSDBF. (j) MGCN. (k) GTransCD. (l) FCFDA.

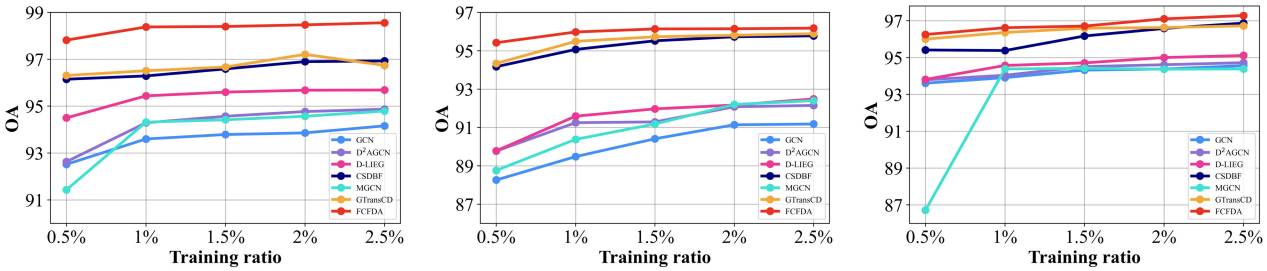
almost 2.93% improvement over the second-place CSDBF. In general, our FCFDA achieves superior performance across several performance indices and shows superiority compared to other methods.

Fig. 6 shows the visual observations of all comparison methods on the River dataset. The visual maps of the GCN, the D^2AGCN, the D-LIEG, the CSDBF, and the MGCN contain a large number of misclassified pixels, especially many FN pixels (red ones). Although the GTransCD has competitive performance, it still has more FP pixels (blue ones) than our FCFDA. The GTransCD and our FCFDA show significantly better visual performance than the above methods but exhibit different visual advantages. The GTransCD has fewer FN pixels, indicating its tendency to detect as many changed pixels as possible. In contrast, our FCFDA generates fewer pseudo-changed pixels (blue ones), demonstrating its ability to ensure the correctness of detecting changed pixels as much as possible.

C. Different Ratios

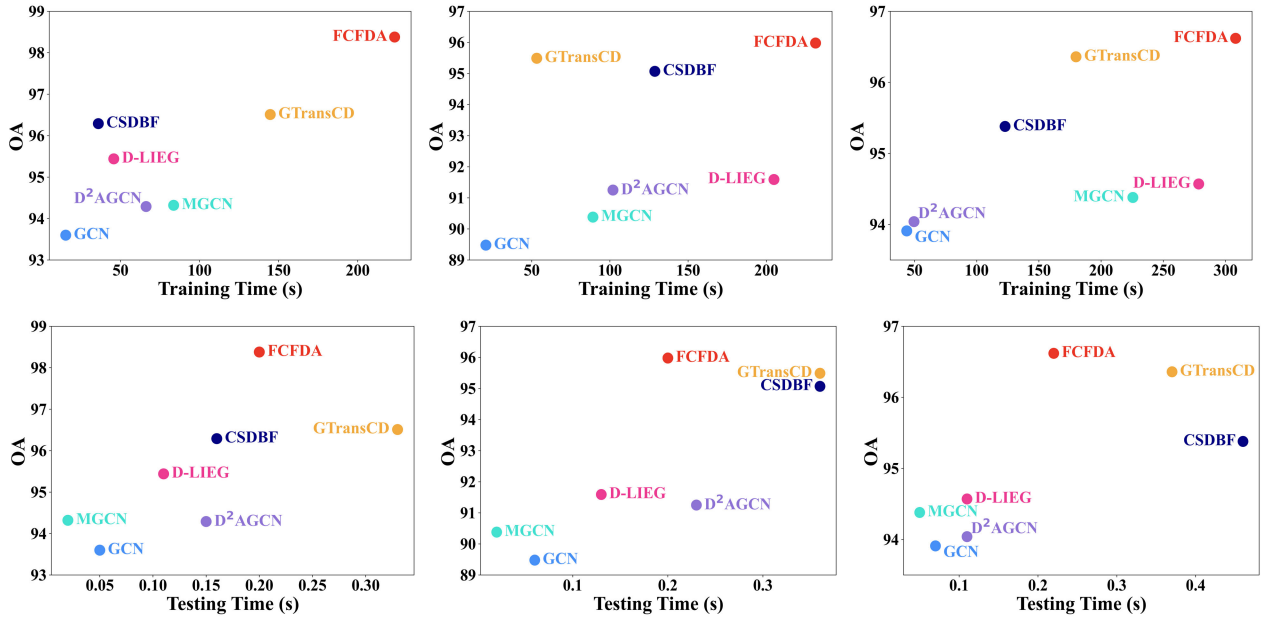
To assess the performance of the proposed FCFDA under various training sample ratios, we randomly select between 0.5% and 2.5% (in increments of 0.5%) of labeled samples from each class for training, using the remaining samples for testing. Fig. 7 illustrates the OAs of all methods on the Farmland, USA, and River datasets across different training sample ratios. First, we observe that the OAs of all methods improve as the number of labeled samples increases. Second, many methods fail to achieve satisfactory accuracies when the training sample ratio is relatively low. For example, the GCN, D^2AGCN, and MGCN perform poorly under the 0.5% training ratio on the Farmland and USA datasets, while the MGCN does not work well on the River dataset under only 0.5% training percentage. In contrast, CSDBF and GTransCD still perform well even under a limited training sample ratio. Notably, the FCFDA consistently achieves the best results across various datasets and different training

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(a) (b) (c)

Fig. 7. CD results of the compared methods with different training ratios on three datasets. (a) Farmland. (b) USA. (c) River.



(a) (b) (c)

Fig. 8. Computational complexity analysis of the compared methods on three datasets. (a) Farmland. (b) USA. (c) River.

sample ratios. Compared to the second-place GTransCD on the Farmland dataset, our FCFDA achieves an average improvement of over 1% across different training sample ratios. In addition, the consistently high performance on these real datasets demonstrates the robustness of our FCFDA to varying numbers of labeled samples.

D. Computational Complexity Analysis

In addition to CD performance, computational complexity is also an important metric for assessing different methods. For this, we record the time required for training and testing. As shown in Fig. 8, the GCN exhibits a short training time on these datasets, while other methods require more training time. For example, D^{2}AGCN achieves better performance but shows longer training time because it increases the computational cost to enhance the difference information during the training stage. GTransCD also improves CD performance but spends more training time mining global information. The FCFDA achieves the best CD performance, but this comes at the expense of computational cost. However, it should be highlighted that our FCFDA only spends comparable testing time but greatly improves CD accuracies, which is superior to other state-of-the-art methods, such as CSDBF and GTransCD, on the USA and River datasets.

E. Hyperparameter Analysis

The sensitivity of the number of hierarchies L is explored in this section. We use the grid search method to determine the number of hierarchies L for various datasets. Note that our method is essentially an FCF, thus L should be taken as a minimum of 2. Therefore, the search range of L is empirically constrained in {2, 3, 4, 5, 6}. The results are displayed in Fig. 9. As shown in the figure, the OA for different datasets shows varying trends as the number of hierarchies L changes. For the Farmland and River datasets, the accuracy increases as L increases from 2 to 4, reaching the highest value at $L = 4$. For the USA dataset, the best accuracy is achieved at $L = 5$, but the accuracy at $L = 4$ is also very high and comparable. To maintain model generalizability, we chose $L = 4$, as it provides a consistent performance improvement across all datasets.

F. Ablation Study

The proposed FCFDA primarily comprises two key modules: the FCF and the DA. To demonstrate the effectiveness

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TABLE IV
ABLATION STUDY ON THE THREE DATASETS

Dataset	FCF	DA	OA(%)	AA(%)	$\kappa \times 100$	F1-score(%)	Precision(%)	Recall(%)
Farmland	×	×	96.81±0.21	96.42±0.18	92.30±0.43	94.56±0.27	93.64±1.56	95.51±1.09
	×	✓	97.77±0.27	97.37±0.53	94.60±0.67	96.17±0.49	95.91±0.20	96.43±1.15
	✓	×	97.70±0.27	97.48±0.27	94.44±0.64	96.07±0.45	95.19±0.66	96.97±0.28
	✓	✓	**98.38±0.17**	**98.07±0.22**	**96.07±0.40**	**97.21±0.29**	**97.08±0.36**	**97.34±0.41**
USA	×	×	95.16±0.12	92.51±0.38	85.98±0.34	89.09±0.27	90.53±1.02	87.70±1.05
	×	✓	95.66±0.24	92.70±0.44	87.30±0.64	90.07±0.48	93.07±1.69	87.30±1.23
	✓	×	95.58±0.15	92.67±0.69	87.08±0.38	89.91±0.31	92.69±2.17	87.36±2.03
	✓	✓	**95.98±0.09**	**93.26±0.50**	**88.26±0.35**	**90.83±0.30**	**93.54±0.75**	**88.29±1.25**
River	×	×	95.78 ±0.27	89.67±1.20	74.92±1.09	77.23±0.97	72.91±2.79	82.27±2.83
	×	✓	96.14±0.31	85.45±3.95	74.31±3.15	76.39±3.03	81.72±5.36	72.52±8.54
	✓	×	95.77±0.30	**90.63±2.03**	75.30±0.50	77.62±0.37	72.16±4.16	**84.42±4.83**
	✓	✓	**96.62±0.08**	86.75±0.54	**77.52±0.15**	**79.35±0.10**	**84.55±1.85**	74.80±1.28

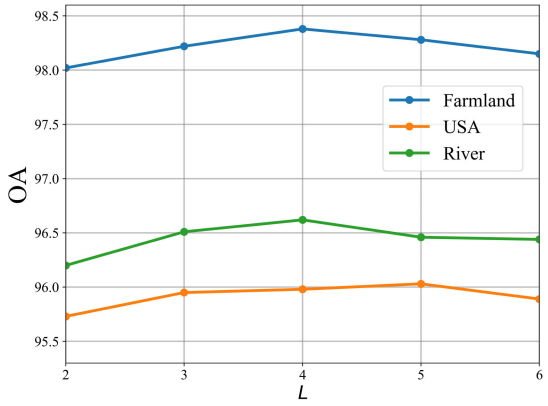


Fig. 9. Parameter analysis of L on three datasets.

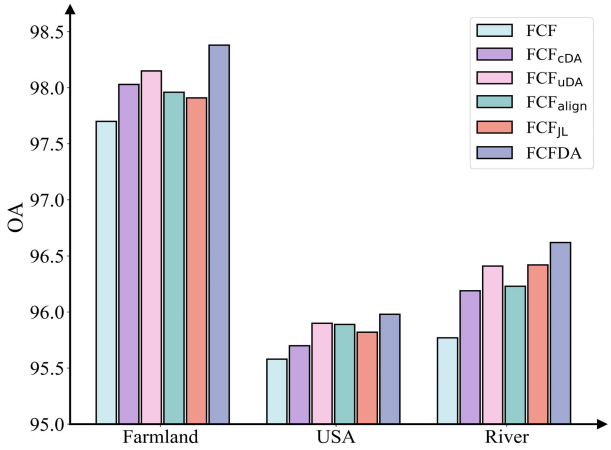


Fig. 10. Ablation study of DA.

of these modules, we fix all other parameters and compare the results when using these modules versus not using them. As shown in Table IV, a significant improvement can be observed with the application of the FCF. This demonstrates that mining different-hierarchy spatial information is beneficial for HSI-CD. Second, comparing the accuracies of the first two variant models shown in Table IV, a significant improvement is observed with the introduction of DA. This demonstrates that aligning the distributions between bitemporal HSIs is capable of significantly reducing pseudo-changes. Moreover, it can be observed that, based on the FCF, the DA further improves model performance. Specifically, OA improves by 0.68%, 0.4%, and 0.85% on the three datasets, respectively. More importantly, Precision improves by 1.89%, 0.85%, and 12.39%, respectively. These improvements are primarily attributed to the tailored DA strategy that alleviates distribution shifts. Overall, the experiments have demonstrated that the FCF and DA effectively improve the performance of HSI-CD.

Second, we also conduct ablation experiments to demonstrate the effectiveness of the proposed DA loss. In our work, the proposed DA loss $\mathcal{L}_{\text{DA}}$ is a combination of the unchanged DA loss $\mathcal{L}_{\text{uDA}}$ and the changed DA loss $\mathcal{L}_{\text{cDA}}$. Therefore, we construct two variant models to evaluate their individual effects. Besides, we also introduce a popular alignment loss [38], represented by the formula $\mathcal{L}_{\text{align}} = (1/N_u)\sum_{i=1}^{N_u} \|\boldsymbol{f}_{1,u_i} - \boldsymbol{f}_{2,u_i}\|_2^2$, for comparison. Three variants are all based on the FCF, but trained with the supervision by $\mathcal{L}_{\text{uDA}}$ (denoted as FCF$_{\text{uDA}}$), $\mathcal{L}_{\text{cDA}}$ (denoted as FCF$_{\text{cDA}}$), and $\mathcal{L}_{\text{align}}$ (denoted as FCF$_{\text{align}}$), respectively. Fig. 10 illustrates the performance comparison on the three datasets. As observed, the baseline model, FCF, performs the worst across all three datasets, indicating its limitations without DA. Models incorporating DA (FCF$_{\text{uDA}}$, FCF$_{\text{cDA}}$, and FCF$_{\text{align}}$) consistently outperform the baseline, highlighting the effectiveness of DA in improving OA. More importantly, our FCFDA achieves the highest accuracy in all datasets, demonstrating the superiority of our proposed DA strategy, which effectively mitigates the impact of distributional differences and significantly enhances detection performance.

Finally, to verify the superiority of the proposed alternating optimization algorithm, we compare the FCFDA with its counterpart model that is trained without using the alternating optimization algorithm. Specifically, this variant model (denoted as FCF$_{\text{JL}}$) is also based on the FCF, but trained with supervision by only a joint loss. Here, the joint loss is a combination of $\mathcal{L}_{\text{DA}}$ and the cross-entropy loss. The results are shown in Fig. 10. As can be seen, compared to the FCFDA,

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FCF_{JL} performs relatively poorly. This indicates that the joint loss has limited effectiveness in achieving proper DA. All these demonstrate the necessity of the proposed alternating optimization algorithm.

IV. Conclusion

This article proposes an FCFDA for HSI CD. First, by generating multilevel graphs to model spatial topologies of bitemporal HSIs hierarchically, the FCFDA fully exploits spatial information among bitemporal HSIs. Second, leveraging the proposed FCF, the FCFDA enhances its ability to extract subtle discriminate features from bitemporal HSIs. These distinguishable subtle features facilitate the detection of both changed and unchanged pixels, demonstrating great fitness for the pixel-wise HSI-CD task. Finally, the proposed DA method helps align the feature space of the bitemporal HSIs, thereby alleviating the challenges posed by distribution shifts in HSI-CD. Comprehensive experiments demonstrate that the proposed FCFDA achieves state-of-the-art performance in terms of both CD accuracy and visualization results.

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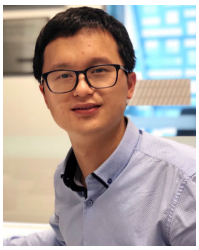
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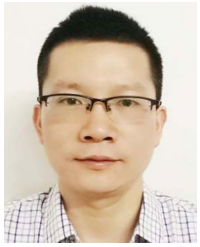
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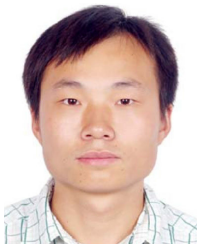
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